



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-	Team No-	Project Title- OmniBank: A Polyglot Microservices Architecture for Distributed Banking Systems
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1. Abstract

Modern financial systems demand high availability, strict ACID compliance, and the ability to process high-volume concurrent transactions without risking data corruption. Traditional monolithic architectures often struggle to scale these operations efficiently as transaction volume and service complexity grow.

OmniBank addresses this by decomposing a banking backend into isolated, purpose-built microservices: a high-performance Payment Gateway (FastAPI), a robust Account & Ledger engine (Java Spring Boot, PostgreSQL) enforcing double-entry ACID transactions, and an event-driven Notification service (Node.js, MongoDB).

To maintain consistency across these independent databases, the system implements the Saga design pattern over RabbitMQ. This event-driven choreography ensures cross-service fund transfers achieve eventual consistency, automatically triggering compensating transactions (refunds) whenever a partial failure occurs.

The architecture further incorporates Redis for API rate-limiting and integrates the pgvector extension within PostgreSQL to enable semantic transaction search — for example, querying “coffee spending” — using HNSW indexing for fast, accurate similarity search.

The complete ecosystem is containerized with Docker Compose and validated through an automated CI/CD pipeline with vulnerability scanning.

Together, these components demonstrate enterprise-grade distributed backend engineering and polyglot persistence within a single, cohesive banking platform.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030265	Bommareddy Haasini Reddy	
2520030285	Arjun Tomar	
2520030558	Rohan Jacob	

3. Frameworks / Platforms / Software Used

Languages: Java 17, Python 3.11, Node.js, SQL

Backend Frameworks: Spring Boot (Spring Data JPA, Spring Web), FastAPI (Pydantic), Express.js

Databases (Polyglot Persistence): PostgreSQL (Relational / ACID), MongoDB (NoSQL Document), Redis (In-Memory Cache)

Vector Database / Search: pgvector extension for PostgreSQL (HNSW Indexing)

Message Broker: RabbitMQ (Saga Pattern Orchestration)

DevOps & Deployment: Docker, Docker Compose

CI/CD & Security: GitHub Actions, Trivy (OWASP Vulnerability Scanning)

Testing & Documentation: JUnit (Java), Pytest (Python), C4 Architecture Diagrams, REST OpenAPI (Swagger)

Guide Name and Signature (remarks if any) _____